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Convertible Bonds: An Introduction

Convertible Bond Research

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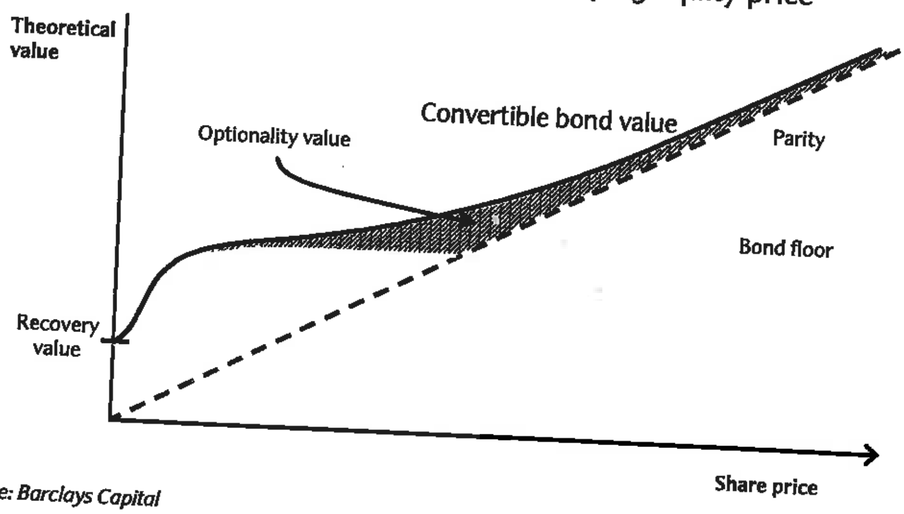
Convertible bonds provide investors with fixed income cash flows, combined with the option to convert into the issuer's equity. This choice gives investors upside participation and downside protection, enabling them to potentially benefit from volatility. Investment decisions can be enhanced by gaining an understanding of the mechanics of the convertible asset class. As such, this report discusses the key components of a convertible bond, together with the investor and issuer rationale. It also introduces a basis for valuation and relative value analysis. Market analysis is provided on a universal level and a typical termsheet is presented and discussed. This document is primarily targeted at those who are relatively new to the convertible product.

What is a convertible bond?

A convertible bond is a fixed income security with a redemption value and, typically, a coupon payment similar to a straight bond. However, in addition the holder has the right to convert the bond into a fixed number of the underlying equity shares. This flexibility results in the theoretical valuation profile illustrated in Figure 1.

Figure 1: Convertible bond value versus underlying equity price



Source: Barclays Capital

Figure 1 illustrates:

- The bond floor, the discounted value of the fixed income cash flows.
- Parity, the value of shares into which the convertible bond would convert.
- The additional value, or optionality, of the convertible bond.

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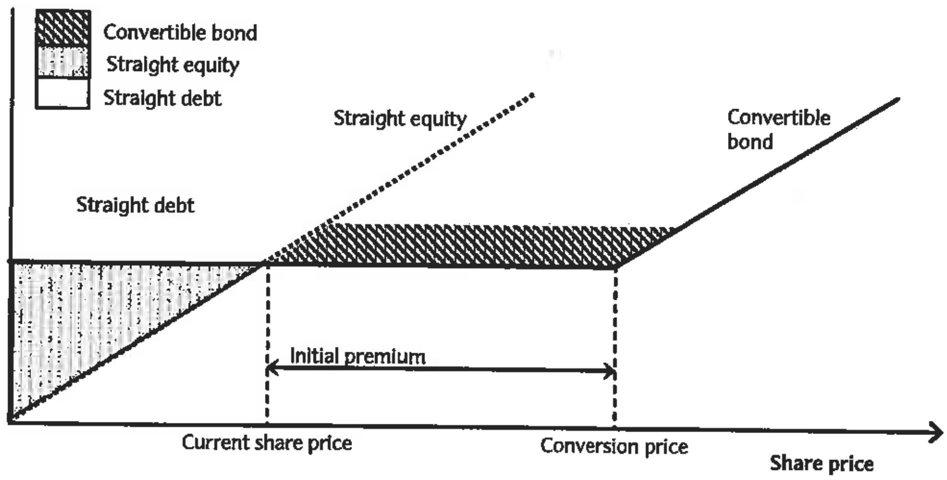
Enabling issuers to achieve financing objectives

- **Reduce interest cost:** The equity option embedded in the convertible allows issuers to monetise the volatility of their stock. This option value provides that a lower yield can be paid as compared to equivalent straight debt. As such, convertible bonds can be an attractive option for companies, especially when straight debt financing costs are high.
- **Generate a premium to the equity sale:** Given the option value embedded within the convertible bond, issuers can offer a convertible bond at an effective share purchase price above the current share price. As such, convertible bonds can fulfil an efficient role in the re-capitalisation of balance sheets. The dynamic of cost savings between a convertible, straight bond and straight equity issue is illustrated in Figure 11.
- **Enhance structuring flexibility:** Embedded call and put options increase the structuring flexibility of the product. These and other clauses can, for example, influence the timing of conversion. Further to this, convertible bonds can attain varying degrees of equity credit from a regulatory perspective.
- **Reach the market quickly:** Convertibles can typically be brought to market with greater expediency than a straight equity issue.
- **Streamline issuance:** Obtaining an agency rating may not be necessary for convertible issuers because the equity option component provides investors with an alternative focus to the credit. Indeed, 44% of the EMEA and 82% of the Asian universe are unrated. Also, convertibles rarely have financial covenants.
- **Diversify investor base:** Convertible bonds have dedicated investors representing both outright (fixed income, equity-oriented, or convertible-specific) and hedged styles, which broadens the investor pool.
- **Provide global exposure:** A number of growth markets have been supported in parallel by the convertible market: for example Asia, the Middle East and South Africa. Also, the three largest corporate Middle Eastern sukuk to date are equity-linked structures, from Nakheel Development ($3.52bn), PCFC Development FCZO ($3.50bn) and Aldar Properties PJSC ($2.53bn).

Figure 11: Relative costs of straight debt, equity and convertible financing



Note: Shading indicates region of lowest-cost financing. Source: Barclays Capital

A

Conversion premium

Conversion premium is the percentage by which market price exceeds parity

The conversion premium is defined as the percentage by which the convertible bond price exceeds parity. At issue, the convertible bond price is par and parity is equal to the reference price, RP, multiplied by the conversion ratio, CR.

As such, the percentage that par is above parity at issue is equivalent to the percentage that CP x CR is above RP x CR; ie, the percentage that the conversion price is above the reference price. Therefore, the conversion premium at issue can be used to calculate the initial conversion price and conversion ratio.

At issue the conversion premium, in conjunction with the reference price, fixes the conversion price and ratio

For the XYZ Corp convertible bond, suppose the VWAP of closing prices was \$10 and that the conversion premium is fixed at 25%. The conversion price would be \$12.5:

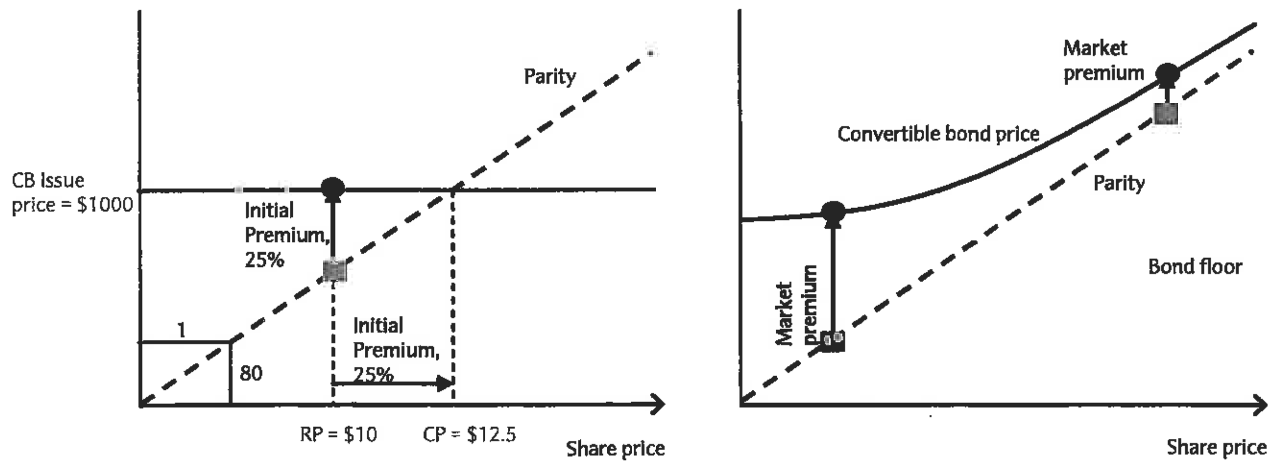
$$CP = RP \times (1 + premium) = \$10 \times (1 + 25\%) = \$12.5$$

As such, the conversion ratio would be 80:

$$CR = \frac{Par}{CP} = \frac{\$1000}{\$12.5} = 80$$

Figure 14 illustrates how the initial (at issue) conversion premium determines the initial conversion price, which in turn determines the conversion ratio. Figure 14 also demonstrates how, over the lifetime of the bond, the conversion premium will typically fall if the share price increases, and will typically rise if the share price decreases.

Figure 14: Conversion details at issue (left) and premiums during the convertible's life (right)



Source: Barclays Capital

B

Calls; issuer's early redemption option

Issuers may have the right to redeem the convertible bond early, after a certain date ...

The convertible bond may be redeemed prior to maturity or 'called', at the issuer's option. The issuer may rationally exercise its call option if the value of the convertible bond would otherwise exceed the call price (typically the principal amount plus accrued). The investor must then choose, during the call notice period, to either redeem the convertible bond at the call price or convert into equity. Typically, a call can be set at any time after the first few years of the bond's life (the average issue from EMEA in 2007 became callable halfway through its tenor). As such, the call feature enables the issuer to limit the potential upside of the bond after those first few years. The issuer's decision to call may be motivated by several factors including whether it wants to extinguish debt from the balance sheet, or whether it needs to refinance at that time, etc.

… but typically subject to a stock trigger level

There is generally a specified share price, or 'trigger' level, which must be breached before the issuer's right to call becomes active. As such, the issuer's call provision is 'soft' and the convertible bond will typically be in the money at the time of the call. For example, a call trigger of 130% would imply that the prevailing share price must be at least 130% of the conversion price for the issuer to be permitted to call.

Calls curtail optionality by inducing early conversion and are seen on the majority of issues

The effect of a soft call is to induce the investor to convert, curtailing optionality once the stock price exceeds the call trigger. Of 2007 EMEA new deals, 80% contained a call feature, of which 94% were soft callable; of 2007 Asian deals, 96% contained a call, of which 96% were soft callable.

C

Puts; holder's early redemption option

Puts limit the investor's exposure to the issuer's credit

Some convertible bonds may be redeemed early, or 'put', by the investor at a fixed price. Exercising this right places a requirement on the issuer to redeem the convertible bond prior to maturity. As such, a put provision limits the length of time the investor is exposed to the credit of the company and protects investors against rising credit spreads or interest rates and against falling equities. Puts would therefore be exercised if the convertible bond value – including the option value – is below the put price. Typically puts can occur only on a small number of discrete dates during the bond's life. Of 2007 EMEA new deals, 33% contained a put feature; of 2007 Asian new deals, 53% contained a put.

D

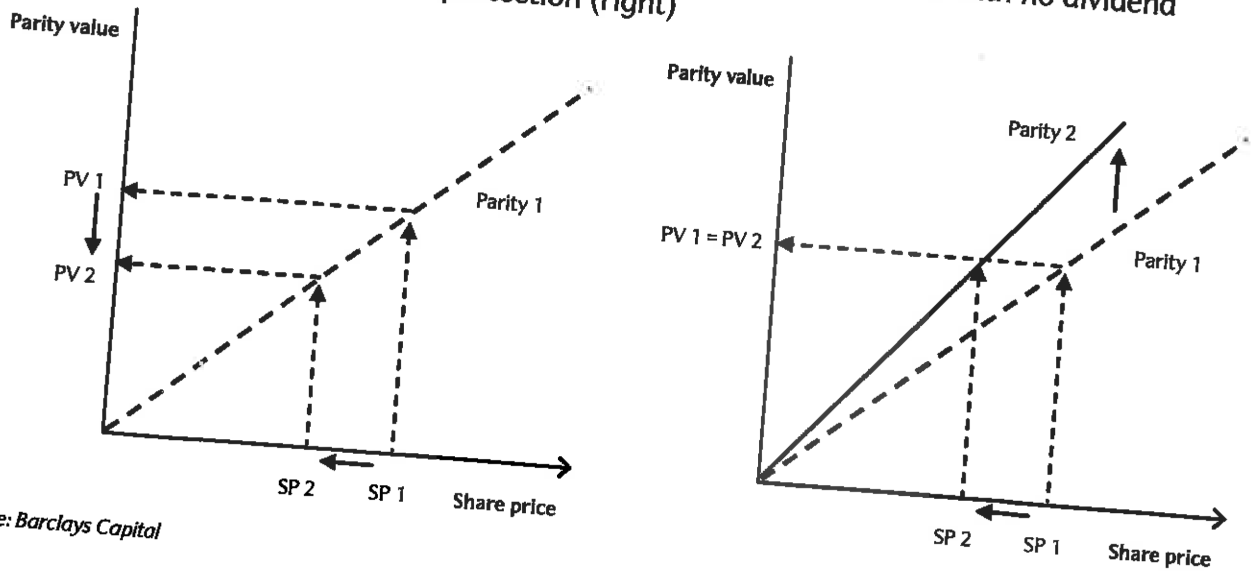
Protection clauses – Change of control and dividends

Protection clauses help maintain optionality

The optionality offered by the convertible bond is a valuable asset. As such, many convertibles include clauses that are designed to offer protection under circumstances where this optionality would be impaired; notably takeovers and dividends.

Standard dividend protection compensates convertible bond holders for any dividend paid above certain threshold levels, which are typically based on current dividend levels or yields. These adjustments are usually implemented by reducing the conversion price (increasing the conversion ratio) in proportion to the size of the dividend above these threshold levels. This serves to protect the optionality of the convertible bond by maintaining (or limiting the fall in) parity across the dividend date. This is illustrated in Figure 15.

Figure 15: Impact of a large dividend upon parity for a convertible bond with no dividend protection (left) and full dividend protection (right)



urce: Barclays Capital

Compensation may also be provided in the event of a change of control

Similarly, takeover protection recompenses for the optionality, which could be lost following a change of control. If the takeover is all equity based then the conversion right of the convertible bond will typically transfer from the acquired stock to the acquirer's stock. In a cash takeover, however, cash has zero volatility, so the optionality embedded in the convertible bond could be lost. Takeover protection often provides compensation for this by means of a downward conversion price adjustment. This is intended to approximately preserve the optionality. Depending on subsequent market conditions, it may under or over-compensate investors, leading to potential trading opportunities around M&A events.

Other structures

The convertible bond landscape is dynamic and evolving

The structure and discussion outlined thus far is for a vanilla convertible. There are a host of variations on this, which will have a marked effect upon valuation. Some examples, which highlight the variety and flexibility afforded by the convertible bond market, include:

- Accreting redemption
  - Instead of, or in addition to paying a coupon, some convertible bonds are structured with an accreting redemption value. However, this yield is only realised if the bond is redeemed.
- Pre-IPO convertibles
  - These may be issued by a company that does not have a public listing at the time of issue. If a public offering of shares occurs before the maturity date, then a percentage of the equity in the offering will be reserved for the convertible bond investors. In some instances, investors would receive additional yield should no IPO occur during the lifetime of the bond. Alternatively a straight bond may be issued, which morphs into a convertible upon the IPO.
- Exchangeable bonds
  - The underlying stock is from a different company to that of the issuer. As such, the credit exposure relates to a different entity to that of the volatility and share price exposure. These may be issued to divest or monetise the volatility of equity stakes.
- Hybrid/preferred convertible securities
  - A subordinated security, which is convertible into the shares of the issuing company. Typical structures may be non-callable for an initial period, long-dated or perpetual, and give the issuer the right to defer coupons or settle them in shares.
- Mandatory convertible securities
  - At maturity the holder must convert the convertible bond into shares. As such, there is no hard bond floor and the equity sensitivity of the instrument is greater than a conventional convertible bond. The conversion ratio at maturity is typically dependant on the final share price, which preserves some degree of asymmetry of payoffs.
- Convertible sukuk
  - A convertible bond based on the principles of Shariah and Islamic finance. In many other respects, the features and embedded options of convertible sukuk are similar to those of conventional convertibles.

The convertible lexicon

Bond floor: The present value of the fixed income cash flows of the security (redemption value plus coupons).

Call: The right of the issuer to redeem the security prior to maturity.

Call protection: A period of time from issue during which the issuer cannot call the bond; typically a few years.

Call trigger: Stock price level that must be reached before the convertible bond can be called. For example, a 130% price trigger would imply that the share price must reach at least 130% of the conversion price before the security can be called by the issuer.

Conversion period: The period during which the security can be converted, typically from soon after issuance until a few days before maturity.

Conversion premium: The amount by which the security price exceeds parity, usually expressed as a percentage of parity. At issue, its price is equal to par. As such, the conversion premium at issue, or 'initial premium' reflects the amount by which the underlying share price must rise to make conversion viable. During the convertible's life, the conversion premium is normally termed 'market premium'.

Conversion price: The share price at which parity equals par, equal to the par value of the security divided by the conversion ratio. Hence, the effective price at which initial purchasers would ultimately be buying the underlying shares, if they convert.

Conversion ratio: The number of shares an investor will receive upon conversion of the security, equal to the par value of the security divided by the conversion price.

Delta: The change in price of the security for a unit change in the underlying share price. Also the equity hedge ratio.

Gamma: The expected change in delta for a unit change in the underlying share price.

Greenshoe or over-allotment option: The right of the underwriter to issue a further amount of the security, typically up to a few weeks after the issue date.

Implied volatility: A metric for relative value comparison. Given a valuation model, a complete set of assumed input parameters will generate a theoretical value. The implied volatility is the volatility assumption, which when used in conjunction with the other assumed parameters will generate the observed market price. If the other parameters are correctly specified, it can therefore be read as the volatility assumption that the market is employing.

Increase option: The right of the issuer to issue a further amount of the security on the day of issue.

Offering circular: A marketing document detailing the security, including terms, issuer, and objectives (also known as the prospectus).

Option adjusted spread: The security's spread over the risk-free rate minus that component of the spread that is attributable to the embedded options.

Parity: The value of the shares a holder would receive upon conversion. Hence, share price multiplied by the conversion ratio, usually expressed as a percentage of par.

Prospectus: See offering circular above.

Put: The right of the investor to require the issuer to redeem the security prior to maturity, typically at a fixed price and on one or more specific dates.